

Write the roles of a 7 members team, named team Zeta ,for website development for Online learning Portal project using devops concepts. Also write the sprint and scrum and the roles of the members. Also provide the downloadable scrum dashboard for this using excel

Team Zeta (7 Members) – Online Learning Portal using DevOps

Member	Role	Responsibilities
Member 1	Scrum Master	Conduct Daily Standups,Sprint Planning,Sprint Reviews, Sprint Retrospectives,Risk Management,Progress Tracking
Member 2	Product Owner	Responsibilities • Gather requirements • Create user stories • Prioritize backlog • Stakeholder communication
Member 3	DevOps Engineer	Application Architecture Database Design Security Design UML Diagrams
Member 4	Frontend Developer	Student Module • Login Page • Dashboard • Attendance Page • Results Page • Assignment Page Faculty Module • Login • Attendance Entry • Marks Upload • Assignment Upload Parent Module

Member	Role	Responsibilities
		- Login - Student Progress Tracking - Attendance Monitoring GitHub Responsibilities - Repository Setup - Branch Management - Pull Requests - Code Reviews
Member 5	Backend Developer	Database Tables Students - StudentID - Name - Course - Semester Faculty - FacultyID - Name - Department Parents - ParentID - StudentID - ContactNumber Attendance - AttendanceID - StudentID - Subject

Member	Role	Responsibilities
		- Percentage Results - ResultID - StudentID - Semester - GPA APIs Student APIs Faculty APIs Parent APIs
Member 6	QA/Test Engineer	Functional Testing - Student Login - Faculty Login - Parent Login - Attendance Module - Results Module - Assignment Module Automation Testing - Selenium - Postman API Testing
Member 7	Cloud & Release Engineer	DevOps Tools - Git - GitHub - Jenkins - Docker - Kubernetes

Member	Role	Responsibilities
		- Prometheus - Grafana Cloud Deployment - AWS EC2 - Azure VM Monitoring - Application Monitoring - Server Monitoring - Log Management Security - SSL Certificates - Role-Based Access Control - Backup Management

Project: Online Learning Portal

Objective

Develop a web-based system for Online Learning Portal assets such as laptops, desktops, printers, software licenses, and network devices.

Scrum Roles

Product Owner

- Maintains Product Backlog
- Prioritizes user stories
- Defines acceptance criteria

Scrum Master

- Conducts Daily Scrum
- Facilitates Sprint Planning

- Removes impediments
- Ensures Scrum practices are followed

Development Team

- Designs, develops, tests, and deploys the application
- Delivers potentially shippable increments every sprint

Sprint Plan

Sprint Plan

Sprint 1 (Weeks 1–2)

Goal

Requirement Analysis & Environment Setup

Tasks

- Requirement Gathering
- User Stories
- Architecture Design
- GitHub Setup
- Jenkins Installation

Deliverables

- Product Backlog
- Architecture Document

Sprint 2 (Weeks 3–4)

Goal

Authentication & User Management

Tasks

- Student Login
- Faculty Login
- Parent Login

- Role-Based Access

Deliverables

- Authentication Module \

Sprint 3 (Weeks 5–6)

Goal

Academic Management

Tasks

- Attendance Module
- Assignment Module
- Results Module
- Notifications

Deliverables

- Academic Management System

Sprint 4 (Weeks 7–8)

Goal

DevOps & Deployment

Tasks

- Docker Containerization
- Jenkins Pipeline
- Automated Testing
- Cloud Deployment
- Monitoring Setup

Deliverables

- Production Deployment
- Monitoring Dashboard

Daily Scrum (15 Minutes)

Each member answers:

1. What did I complete yesterday?

2. What will I work on today?
3. Are there any blockers?

Example:

Member	Yesterday	Today	Blocker
Frontend Developer	Login UI	Dashboard UI	None
Backend Developer	Authentication API	Asset API	None
QA Engineer	Login Testing	API Testing	Waiting for API

DevOps Tools

Activity	Tool
Version Control	Git
Repository	GitHub
CI/CD	Jenkins / GitHub Actions
Containerization	Docker
Monitoring	Prometheus, Grafana
Issue Tracking	Jira
Cloud Deployment	AWS / Azure

Team_Zeta_Online_Learning_Portal_Dashboard.xlsx

[illegible]

Scrum DashboardTeam RolesSprint Plan

The workbook contains:

- Scrum Dashboard
- Team Roles
- Sprint Plan

Responsibilities of each member in detail

Team Zeta – Detailed Roles and Responsibilities

Project: Online Learning Portal using DevOps

Modules: Student Login, Faculty Login, Parent Login, Course Management, Attendance Management, Assignment Management, Results Management, Notifications

1. Member 1 – Scrum Master

Primary Responsibility

Ensure the team follows Scrum practices and completes sprint goals on time.

Detailed Responsibilities

Sprint Management

- Conduct Sprint Planning meetings.
- Organize Daily Scrum meetings.
- Schedule Sprint Review and Retrospective meetings.
- Monitor sprint progress using Scrum Dashboard.

Team Coordination

- Coordinate among developers, testers, and DevOps engineers.
- Ensure tasks are evenly distributed.
- Track blockers and project risks.

Project Tracking

- Monitor sprint velocity and burndown charts.
- Ensure user stories are completed within sprint deadlines.
- Update Jira/Trello boards regularly.

Example Tasks

- Conduct daily stand-up meetings.
- Review sprint backlog.
- Track unresolved issues.
- Communicate project status to stakeholders.

Deliverables

- Sprint Plans
- Sprint Reports

- Burndown Charts

2. Member 2 – Product Owner

Primary Responsibility

Define system requirements and prioritize features based on stakeholder needs.

Detailed Responsibilities

Requirements Gathering

- Meet with students, faculty, parents, and administrators.
- Collect requirements for online learning activities.
- Define portal objectives.

Product Backlog Management

Create and Prioritize User Stories

- Student Registration & Login
- Faculty Login
- Parent Login
- Attendance Management
- Assignment Submission
- Result Viewing
- Course Management
- Notifications

Example User Story

As a Student, I want to view my attendance online so that I can track my academic progress.

Acceptance Criteria

- Student must log in successfully.
- Attendance percentage should be displayed.
- Attendance data should be updated automatically.
- Semester-wise attendance should be available.

Deliverables

- Product Backlog
- User Stories
- Acceptance Criteria
- Requirement Specification Document

3. Member 3 – DevOps Engineer

Primary Responsibility

Automate building, testing, and deployment of the Online Learning Portal.

Detailed Responsibilities

Source Code Management

- Create Git Repository.
- Define Branching Strategy.
- Manage Pull Requests.

CI/CD Pipeline Development

Pipeline Flow

1. Code Commit
2. Build Application
3. Run Automated Tests
4. Create Docker Image
5. Deploy to Cloud Server

Docker Containerization

- Create Dockerfiles.
- Build application containers.
- Manage Docker images.

Automation

- Automate deployments.
- Automate testing.
- Configure GitHub Webhooks.

Monitoring

- Configure Prometheus.
- Create Grafana Dashboards.
- Monitor application health and uptime.

Deliverables

- CI/CD Pipeline
- Docker Containers
- Deployment Scripts
- Monitoring Dashboards

4. Member 4 – Frontend Developer

Primary Responsibility

Develop the user interface of the Online Learning Portal.

Detailed Responsibilities

User Interface Development

Pages to Develop

Student Module

- Login Page
- Student Dashboard
- Attendance Page
- Assignment Page
- Results Page

Faculty Module

- Faculty Login
- Attendance Entry Page
- Assignment Upload Page
- Marks Entry Page

Parent Module

- Parent Login
- Student Progress Dashboard
- Attendance Monitoring Page
- Result Monitoring Page

UI Design

- Create responsive layouts.
- Ensure mobile compatibility.
- Improve user experience.

API Integration

- Connect frontend with backend APIs.
- Display academic information dynamically.

Example Dashboard Features

Student Dashboard

Displays:

- Attendance Percentage
- Assignment Status
- Course Information
- Semester Results

Deliverables

- Web Pages
- Responsive Design
- API Integration

5. Member 5 – Backend Developer

Primary Responsibility

Develop server-side functionality and business logic.

Detailed Responsibilities

Database Design

Students Table

- StudentID
- StudentName
- Email
- Course
- Semester

Faculty Table

- FacultyID
- FacultyName
- Department
- Email

Parents Table

- ParentID
- ParentName
- StudentID
- ContactNumber

Attendance Table

- AttendanceID
- StudentID
- Subject
- AttendancePercentage

Results Table

- ResultID
- StudentID
- Semester
- GPA

Assignments Table

- AssignmentID

- Subject
- DueDate
- SubmissionStatus

API Development

Student APIs

GET /student/profile

GET /student/attendance

GET /student/results

GET /student/assignments

Faculty APIs

POST /attendance

POST /marks

POST /assignments

GET /students

Parent APIs

GET /parent/student-progress

GET /parent/attendance

GET /parent/results

Business Logic

- Attendance Validation
- Assignment Submission Validation
- Result Calculation
- Student Progress Tracking

Security

- Authentication
- Authorization
- Password Encryption
- Session Management

Deliverables

- Database Schema
- REST APIs
- Business Logic Modules

6. Member 6 – QA/Test Engineer

Primary Responsibility

Ensure the Online Learning Portal works correctly and is free from defects.

Detailed Responsibilities

Test Planning

Prepare:

- Test Plan
- Test Cases
- Test Scenarios

Functional Testing

Verify:

- Student Login
- Faculty Login
- Parent Login
- Attendance Management
- Assignment Submission
- Result Viewing
- Notifications

Automation Testing

- Selenium Scripts
- API Testing using Postman

Bug Tracking

Bug ID	Description	Status
BUG001	Student Login Failure	Open
BUG002	Attendance Not Updating	Fixed
BUG003	Assignment Upload Error	Open

Regression Testing

- Verify old features after updates.
- Ensure new code doesn't break existing functionality.

Deliverables

- Test Cases
- Bug Reports
- Test Summary Reports

7. Member 7 – Cloud & Release Engineer

Primary Responsibility

Deploy and maintain the Online Learning Portal on cloud infrastructure.

Detailed Responsibilities

Cloud Setup

Provision:

- Virtual Machines
- Databases
- Storage Services

Using:

- AWS EC2
- Azure Virtual Machines
- Google Cloud VM

Deployment

- Configure Web Servers.

- Deploy Docker Containers.
- Manage Production Releases.

Monitoring

Track:

- CPU Usage
- Memory Usage
- Application Uptime
- Response Time

Backup & Recovery

- Database Backups
- Disaster Recovery Planning
- Restore Procedures

Security Management

- SSL Certificates
- Firewall Rules
- Access Control Management

Deliverables

- Cloud Infrastructure
- Production Environment
- Monitoring Reports
- Backup Strategy

Responsibility Matrix (RACI)

Activity	SM	PO	DevOps	Frontend	Backend	QA	Cloud
Requirements	C	A/R	I	I	I	I	I
Sprint Planning	A/R	R	C	C	C	C	C

Activity	SM	PO	DevOps	Frontend	Backend	QA	Cloud
UI Development	I	C	I	A/R	C	C	I
API Development	I	C	I	C	A/R	C	I
Testing	I	I	C	C	C	A/R	I
CI/CD Pipeline	I	I	A/R	C	C	C	C
Cloud Deployment	I	I	C	I	C	I	A/R
Monitoring	I	I	R	I	I	I	A/R

Legend:

- **A** = Accountable
- **R** = Responsible
- **C** = Consulted
- **I** = Informed

This structure closely matches how a real-world Agile DevOps team would develop and deploy an IT Asset Management web application over 4 sprints.